CONNECTED HOME ENERGY FORUM · BERLIN, 26 OCTOBER 2026

Electrification without interoperability

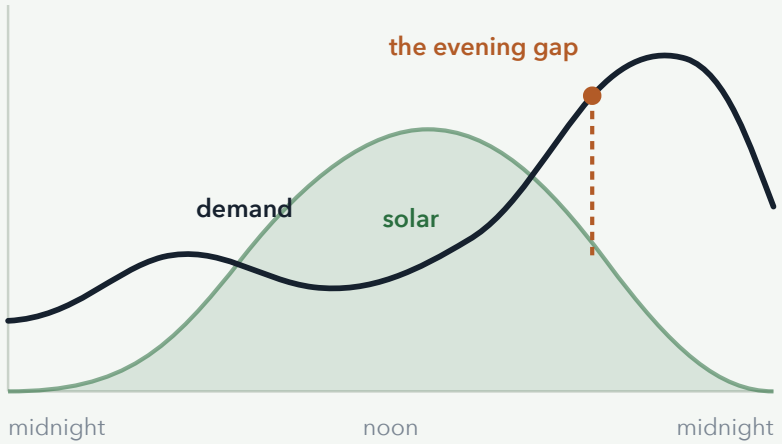
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Electrification creates two problems. Today we are solving one.

BALANCE THE GRID

The entire grid, hour by hour

Supply must match demand, always.

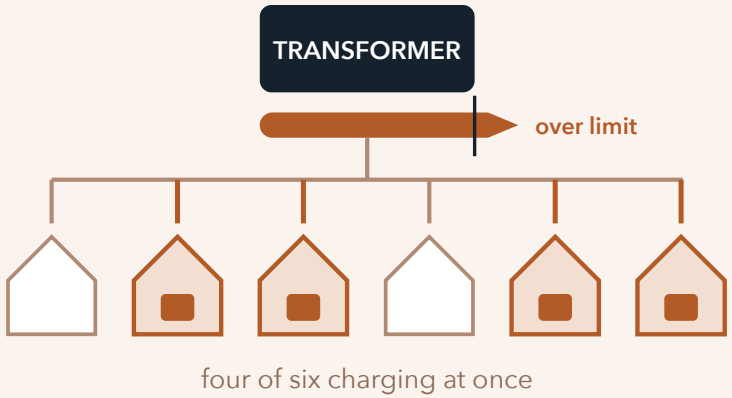


An economics problem. Move energy in time.

PROTECT THE TRANSFORMER

One block, right now

The transformers and wires have a fixed capacity.



four of six charging at once

A physics problem. Cap power right now.

Balancing the grid is an **economics** problem. Protecting the transformer is a **physics** problem. **Both land on the customer's bill.** Peak we cannot shift is generation and transmission somebody has to build; a transformer we cannot protect is a street somebody has to dig up.

One signal cannot do both jobs

	A DYNAMIC PRICE	A PER-CUSTOMER POWER LIMIT
What it is	What energy costs right now	How much power this connection may draw right now
Unit	The entire grid	One service connection
What it solves	Balancing the grid	Protecting the transformer
What it does	Moves energy across time	Caps power, immediately
Without it	Nobody has a reason to shift	Nothing stops everyone shifting at once

A price is an incentive shared by everyone on the same asset. It cannot stop five houses on one transformer responding to the same number at the same moment. **A price is an incentive over a group. A limit is an obligation on one connection.**

WHAT THE DISTRIBUTION NETWORK IS ALREADY DOING

We are already experiencing the transformer problem

Voltage rises on a sunny afternoon, and inverters and chargers trip off to protect themselves. The customer sees a fault, not a grid condition.

Connections are refused, queued or derated for heat pumps and chargers, because nobody can promise what a house will draw at the worst moment.

Capacity is managed by curtailment and by waiting, because there is no signal that tells a connection how much it may take right now.

> Every one of these is about protecting the transformer. **No signal in use today carries an answer to them.**

The good news: three standards already carry a price

	CARRIES A PRICE TODAY
OpenADR 3	Utility to the customer's own equipment, over the Internet
Matter	The Commodity Price cluster, inside the home
EEBUS	Time of Use Tariff: "the absolute price SHALL be included", currency per Wh

Tomorrow you will hear from Matter and from OpenADR on exactly this. **This half of the problem has an answer.**

RUNNING TODAY

A free, public price server, no account required

Real hourly prices, live right now, for two large utilities across their distribution circuits and substations, plus prices computed from published rate schedules.

Hourly marginal carbon on the same server, in the same shape. A greenhouse-gas number is price-shaped: a device optimizes against it exactly as it does a price.

No authentication, no contract, no enrollment. Open-source home automation receives it and orchestrates the house against it.

> `price.grid-coordination.energy/openadr3/3.1.0/programs?limit=5`
> Try it in a browser during the break.

Three standards already define a capacity limit

	AN IMPORT AND EXPORT CAPACITY LIMIT FOR THE SITE
OpenADR 3	A **dynamic operating envelope**: a schedule of limits, with a default the device falls back to when communications are lost
EEBUS	**Power Envelope**, defined at "the building's grid connection point", import and export, time-varying, with a declared fallback
Matter	A power threshold is defined in the Application Cluster specification

We know how to do this. Now we have to require it.

Nothing obliges a device to receive one. Not a standard, not a program, not a procurement.

The limit must be on the connection, not the device. Today's programs cap the appliances you enrolled, which leaves the transformer exposed to everything else in the house.

> **No certification program tests whether a device can receive a limit.**

OF 376 PRODUCTS REGISTERED UNDER THE EU CODE OF CONDUCT

About 7% built it into the appliance

The **EU Code of Conduct on interoperability of energy smart appliances** declares the communication layer out of scope.

The Commission's own evaluation counted **376 registered compliant products**.

A vendor proprietary protocol to their own cloud	92.57%
Built into the appliance	**6.90%**
Dongle	0.53%

These are the Commission's own numbers, about its own rule. **Require an open protocol on the appliance itself. Tunnel it back to your own cloud as well if you want to, but the cloud cannot be the thing that satisfies the requirement.** Matter is local by construction, which is what makes tomorrow's session worth your attention.

A CURRENT EVSE CERTIFICATION PROGRAM, SCORED AGAINST SIX TESTS

It certifies dispatch, and neither signal

TEST	VERDICT	IN THE RELEASED SPECIFICATION
1. An objective, not operation	**Not met**	A Charging Schedule Function with default and **dispatch** schedules
2. Open all the way to the appliance	**Met**	Mapped onto an open protocol that terminates on the charger
3. A server the customer chooses	**Not met**	"server address" and "endpoint": zero occurrences
4. Direct from the utility	**Not met**	Aggregator mediated throughout
5. A limit that holds locally	**Not met**	"capacity limit", "import limit", "export limit": zero occurrences
6. The customer who cannot respond	**Met**	A local override, which expires every time the vehicle is unplugged

The customer agrees to the capacity limit, and is paid for it

Nobody is telling anyone what they may draw in their own house. A capacity limit is something a customer opts into: accept a cap, and get connected sooner, or cheaper, or get paid for the flexibility.

Regulators are already ordering it. California has directed its two largest utilities to establish a standardized tariffed flexible connection.

The commercial model is not the gap. The customer signs the agreement, and then nothing tells the equipment in the house what was agreed.

> **The carrier is the gap.** A limit that lives only in a contract is enforced by the customer, by hand, or not at all.

THE ASK

What you can do next

If you write the standard: make both signals conformance-testable rather than optional. A device type that may read a price is a device type that will not.

If you build the product: receive both. Read a price, respect a capacity limit, and let the customer point the server somewhere else.

If you run the network or the service: publish the limit, and ask for both in your next supplier requirement.

Why wait for a regulator to mandate it, and then have to react? Nobody here has to invent anything: both signals are already defined in the standards you already build on. **Put them in the next specification you write, and the market will have them inside two years.**

EVERYTHING HERE IS PUBLIC AND FREE

Where to find all of this

The price server	`price.grid-coordination.energy/openadr3/3.1.0/programs?limit=5`
User guide, tutorials, client libraries	`github.com/grid-coordination`
The argument, in full	`grid-coordination.energy`
All the decks	`grid-coordination.energy/presentations/`

No account, no contract, no enrollment on any of it.